

There are a total of six problems. You have to solve all the problems.

Problem 1 (CO1): Finite Automata (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

$$L_1 = \{w : w \text{ contains even number of 0s before the first 1}\}$$

$$L_2 = \{w : \text{no 0 appears after the first 1 in } w\}$$

$$L_3 = \{w : \text{every third symbol of } w \text{ is 1}\}$$

$$L_4 = L_1 \cap L_2 \cap L_3$$

Now solve the following problems.

- Give the state diagram of a DFA that recognizes L_1 . (2 points)
- Give the state diagram of a DFA that recognizes $\overline{L_3}$. [Recall: $\overline{L}$ denotes the complement of the language L i.e., $\overline{L} = \Sigma^* - L$] (3 points)
- If you are to use the “cross product” construction to obtain a DFA that recognizes $L_1 \cap \overline{L_3}$, how many states will it have? (1 point)
- Write four strings of length five in $L_1 \cap L_2$. (1 point)
- Write all strings of length four in L_4 . (1 point)
- Give the state diagram of a DFA that recognizes L_4 using exactly five states. (2 points)

Problem 2 (CO1): Regular Languages (5 points)

Write whether the following statements are true or false. In addition, write the correct statements for the false statements.

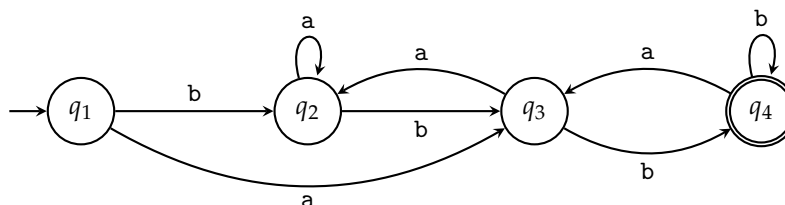
- There exists a language that can be recognized by some NFA with ϵ -transitions but cannot be recognized by any NFA without ϵ -transitions. (1 point)
- If L_1 and L_2 are regular languages, then $L_1 - L_2$ is also a regular language. [Recall: $L_1 - L_2$ contains all strings that are in L_1 but not in L_2] (1 point)
- If every state of an NFA is an accepting state, then it accepts every possible string over its alphabet. (1 point)

Answer the following questions.

- Identify which of the following statements are true. (1 point)
 - $(b^*a^*)^* = (a^* + b^*)^*$
 - $(ab)^*a = a(ba)^*$
- Write the shortest string over $\Sigma = \{0, 1\}$ that is not generated by the regular expression $(1 + 01 + 001)^*(\epsilon + 0 + 00)$. (1 point)

Problem 3 (CO2): Converting FA to Regular Expressions (10 points)

Convert the following finite automata into an equivalent regular expression using the state elimination method. You must eliminate the states in the following order: q_3, q_2, q_4 . Show each step of the process.



Problem 4 (CO2): Converting Regular Expressions to NFA (10 points)

Convert the following regular expression over $\Sigma = \{a, b\}$ into an equivalent NFA. Note that $R_1 + R_2$ is the same as $R_1 \cup R_2$.

$$ba(a + a^*b)^* + (ab)^*a$$

Problem 5 (CO1): Regular Expressions (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

$$L_1 = \{w : w \text{ contains } 011\}$$

$$L_2 = \{w : w \text{ does not contain } 00\}$$

$$L_3 = L_1 \cap L_2$$

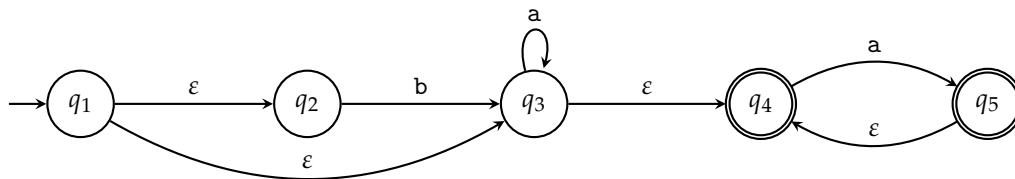
$$L_4 = \{w : \text{every substring of length 3 in } w \text{ contains exactly two 1s}\}$$

Now solve the following problems.

- Give a regular expression that generates L_1 . (2 points)
- Give a regular expression that generates L_2 . (2 points)
- Does there exist a string in L_2^* that contains 00 as a substring? **Explain** your answer. (1 point)
- Give a regular expression that generates L_3 . (2 points)
- Write two strings of length six in L_4 , one starting with 0 and the other starting with 1. (1 point)
- Give a regular expression that generates L_4 . (2 points)

Problem 6 (CO2): Converting NFA to DFA (5 points)

Consider the following NFA:



Now answer the following questions.

- If you convert the given NFA into an equivalent DFA using the subset construction method, **what** is the maximum number of accepting states that the DFA can have? (1 point)
- Write the ϵ -closure of state q_1 in the given NFA. (1 point)
- What** is $\delta(\{q_4\}, a)$ in the given NFA? [Recall: $\delta(\{q\}, a)$ is the set of states in which the NFA transitions when it is in state q and receives input a .] (1 point)
- What** is $\delta(\{q_2, q_3, q_4\}, a)$ in the given NFA? (2 points)

After completing the exam, please indicate where you stand on the smiley face spectrum.

